# Calculation questions – 4.3 Particle model of matter

AQA GCSE Physics (8463) Higher · Paper 1 · 12 questions · 41 marks · answers at the back

## Equations for this topic

| Equation | Symbols and units | Exam |
|---|---|---|
| ρ = m/V | ρ (kg/m³), m (kg), V (m³) | Learn it |
| ΔE = mcΔθ | ΔE (J), m (kg), c (J/kg °C), Δθ (°C) | On sheet |
| E = mL | E (J), m (kg), L (J/kg) | On sheet |
| pV = constant (triple) | p (Pa), V (m³), constant temperature | On sheet |
| E = Pt | E (J), P (W), t (s) | Learn it |

Write the equation, substitute, then rearrange. Convert to base units first. Give a unit and 2–3 significant figures. Take g = 9.8 N/kg.

## Grade 4–5

**1** A block has a mass of 2.7 kg and a volume of 0.0010 m³. Calculate its density. [2 marks]

Answer: _____ Unit: _____

**2** Specific latent heat of fusion of ice = 334 000 J/kg. Calculate the energy needed to melt 0.50 kg of ice at 0 °C. [2 marks]

Answer: _____ Unit: _____

**3** A stone has density 2500 kg/m³ and volume 0.0040 m³. Calculate its mass. [2 marks]

Answer: _____ Unit: _____

## Grade 6–7

**4** A metal cube has sides of 5.0 cm and a mass of 0.95 kg. Calculate its density in kg/m³. [3 marks]

Answer: _____ Unit: _____

**5** Specific latent heat of vaporisation of water = $2.26 \times 10^6$ J/kg. Calculate the energy needed to boil away 200 g of water at 100 °C. [3 marks]

Answer: _____ Unit: _____

**6** A 2.0 kW kettle keeps water boiling for 3.0 minutes. Assuming no energy is wasted, calculate the mass of water turned to steam. Use the value of L from Q5. [4 marks]

Answer: _____ Unit: _____

**7** (Triple) A gas at constant temperature has a pressure of 100 kPa and volume 0.030 m³. It is compressed to 0.020 m³. Calculate the new pressure. [3 marks]

Answer: _____ Unit: _____

**8** A stone of mass 64.8 g is lowered into a measuring cylinder. The water level rises from 50 cm³ to 74 cm³. Calculate the density in g/cm³ and in kg/m³. [3 marks]

Answer: _____ Unit: _____

## Grade 8–9

**9** Calculate the total energy needed to turn 0.20 kg of ice at −10 °C into water at 20 °C. c(ice) = 2100 J/kg °C, c(water) = 4200 J/kg °C, L(fusion) = 334 000 J/kg. [6 marks]

Answer: _____ Unit: _____

**10** A 50 W heater melts 0.15 kg of a solid. The temperature stays constant for 6.0 minutes while it melts. (a) Calculate the specific latent heat of fusion. (b) Explain whether your value is likely to be too high or too low. [5 marks]

Answer: _____ Unit: _____

**11** (Triple) A bicycle pump holds 120 cm³ of air at 1.0 × 10⁵ Pa. It is compressed slowly to 40 cm³. (a) Calculate the new pressure. (b) Explain why the air gets warmer if the pump is pushed in quickly. [5 marks]

Answer: _____ Unit: _____

**12** (Triple) A bubble of volume 2.0 cm³ is released at the bottom of a lake where the pressure is 2.5 × 10⁵ Pa. It rises to the surface, where the pressure is 1.0 × 10⁵ Pa. Assuming constant temperature, calculate its volume at the surface. [3 marks]

Answer: _____ Unit: _____

# Answers – 4.3 Particle model of matter

Mark yourself honestly: a mark for the right equation, a mark for substitution, a mark for the answer with its unit.

| Q | Working | Answer |
|---|---|---|
| 1 | 2.7 ÷ 0.0010 | **2700 kg/m³** |
| 2 | 0.50 × 334 000 | **167 000 J** |
| 3 | 2500 × 0.0040 | **10 kg** |
| 4 | $V = 0.050^3 = 1.25 \times 10^{-4}$ m³; 0.95 ÷ $1.25 \times 10^{-4}$ | **7600 kg/m³** |
| 5 | $0.200 \times 2.26 \times 10^6$ | **$4.52 \times 10^5$ J** |
| 6 | E = 2000 × 180 = 360 000 J; m = 360 000 ÷ $2.26 \times 10^6$ | **0.16 kg** |
| 7 | 100 × 0.030 = p × 0.020 | **150 kPa** |
| 8 | V = 74 − 50 = 24 cm³; 64.8 ÷ 24 | **2.7 g/cm³ = 2700 kg/m³** |
| 9 | warm ice 0.20 × 2100 × 10 = 4200 J; melt 0.20 × 334 000 = 66 800 J; warm water 0.20 × 4200 × 20 = 16 800 J | **87 800 J** |
| 10 | E = 50 × 360 = 18 000 J; L = 18 000 ÷ 0.15. Too high: some heater energy goes to the surroundings, so the substance absorbed less than 18 000 J | **(a) 120 000 J/kg (b) too high** |
| 11 | $1.0 \times 10^5 \times 120 = p \times 40$. Work is done on the gas, increasing the particles' kinetic energy, so temperature rises | **(a) $3.0 \times 10^5$ Pa** |
| 12 | $2.5 \times 10^5 \times 2.0 = 1.0 \times 10^5 \times V$ | **5.0 cm³** |

**Common mistakes.** Cubing before converting cm to m (Q4: 5.0 cm = 0.050 m); missing a stage in multi-stage heating (Q9); using a temperature change in E = mL (there isn't one).